

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1592

Corson 138 kV
September 02, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1592. The Interconnection Customer proposes a 122.9 MW solar plus storage facility interconnecting at the Corson 138 kV substation in the Virginia area. The request was received on October 10, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 416 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,627,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
SPING STREET-CHIPOLA 138 KV	MITCHELL DAM-SKAGGS ISLAND JUNCTION 2 345 KV line	109.3	234	7.2
JETERSVILLE-CREWE 138 KV	BEAR-CLOVER 345 KV	107.3	279	55.8
CORSON-CREWE 138 KV	CLOVER-BEAR 345 KV	106.1	303	55.8
FRITCH-BIG HANAFORD 138 KV	MITCHELL DAM-SKAGGS ISLAND JUNCTION 2 345 KV	104.7	854	6.2
CHANCELLOR ROW-NATCHEZ 138 KV	SKAGGS ISLAND JCT 2-MITCHELL DAM 345 KV	104.2	508	5.4
LAKESIDE-NATCHEZ 138 KV	MITCHELL DAM-SKAGGS ISLAND JUNCTION 2 345 KV	103.7	531	5.4
FRITCH-LAKESIDE 138 KV	SKAGGS ISLAND JCT 2-MITCHELL DAM 345 KV line	102.4	607	5.4
HULL ST RD-HARVARD 138 KV	MITCHELL DAM-SKAGGS ISLAND JCT 2 345 KV line	102.0	574	7.4
SECO-HARVARD 138 KV	MITCHELL DAM-SKAGGS ISLAND JUNCTION2 345 KV	101.9	524	7.4
WING LANE-SECO 138 KV	MITCHELL DAM SUBSTATION-SKAGGS ISLAND JCT 2 345 KV	101.6	413	7.4
SELMA-BLUFF VIEW 138 KV	CLOVER-BEAR 345 KV	101.1	272	55.8
JETERSVILLE-BLUFF VIEW 138 KV Line	BEAR-CLOVER 345 KV	100.8	286	55.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CHIPOLA-SPRING ST 138 KV	Reconductor line	\$3,411,000	3% (\$87,000)
JETERSVILLE-CREWE 138 KV	Reconductor line	\$3,577,000	66% (\$2,345,000)
CORSON-CREWE 138 KV	Reconductor line	\$3,024,000	66% (\$1,982,000)
BIG HANAFORD-FRITCH 138 KV	Reconductor line	\$2,213,000	1% (\$16,000)
CHANCELLOR ROW-NATCHEZ 138 KV	Reconductor line	\$1,391,000	1% (\$14,000)
LAKESIDE-NATCHEZ 138 KV	Reconductor line	\$1,190,000	1% (\$12,000)
LAKESIDE-FRITCH 138 KV	Reconductor line	\$1,462,000	1% (\$15,000)
HULL ST RD-HARVARD 138 KV	Reconductor line	\$595,000	2% (\$9,000)
SECO-HARVARD 138 KV	Reconductor line	\$2,125,000	2% (\$32,000)
WING LANE-SECO 138 KV	Reconductor line	\$550,000	2% (\$8,000)
SELMA-BLUFF VIEW 138 KV	Reconductor line	\$1,024,000	66% (\$671,000)
JETERSVILLE-BLUFF VIEW 138 KV	Reconductor line	\$3,716,000	66% (\$2,436,000)
	Total allocated to project		\$7,627,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.